

IMPACT OF THE 2020 EUCAST « I » DEFINITION ON ANTIPSEUDOMONAL TREATMENT CHOICES IN BELGIAN HOSPITALS : A MULTICENTRIC STUDY

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Introduction

- In 2020, EUCAST redefined the “I” category in antimicrobial susceptibility testing (AST) to “Susceptible, at increased exposure”, aiming to promote narrow-spectrum antibiotic use.
- By July 2022, Belgian laboratories adopted this change, classifying first-line antipseudomonal antibiotics as “I”, while meropenem remained “S”.

Objectives

- Assess the impact of these changes on antibiotic selection for *Pseudomonas aeruginosa* infections in Belgian hospitals.
- Identifying factors influencing meropenem prescriptions, as healthcare practitioners may now opt for broader-spectrum antibiotics in treatment.

Methods

Design

- Multicentric retrospective observational study
- Belgian acute care hospitals - voluntary basis

Population

- Hospitalized adults treated for wild-type monomicrobial *Pseudomonas aeruginosa* infections:
- Patients were included in the study only if the isolate was identified as

	Before 2020 update	After 2020 update
Ceftazidime	S	I
Piperacillin/tazobactam	S	I
Ciprofloxacin	S	I
Cefepime	S	I
Meropenem	S	S

Data collection

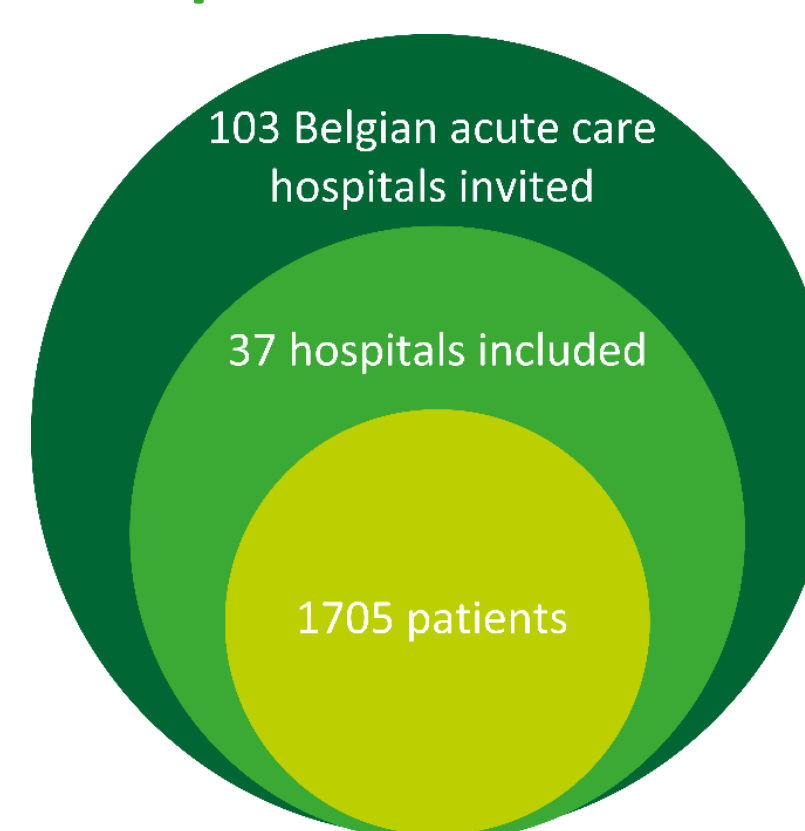
November 2023 - June 2024 

- Hospital data (characteristics, AST reporting strategies, information campaigns)
- Patient data (demographic, clinical, microbiological, and therapy details)

Analysis

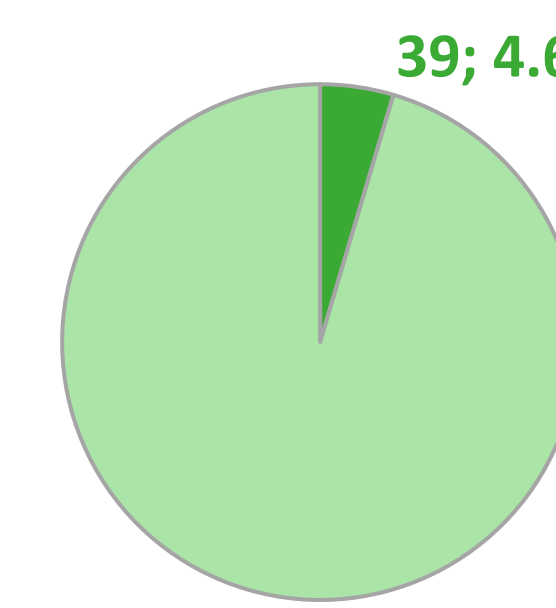
Results

Participation



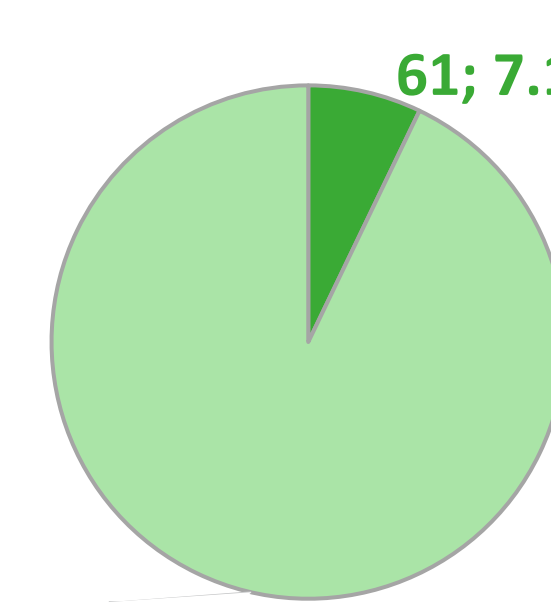
Pre-shift

850 patients treated after AST



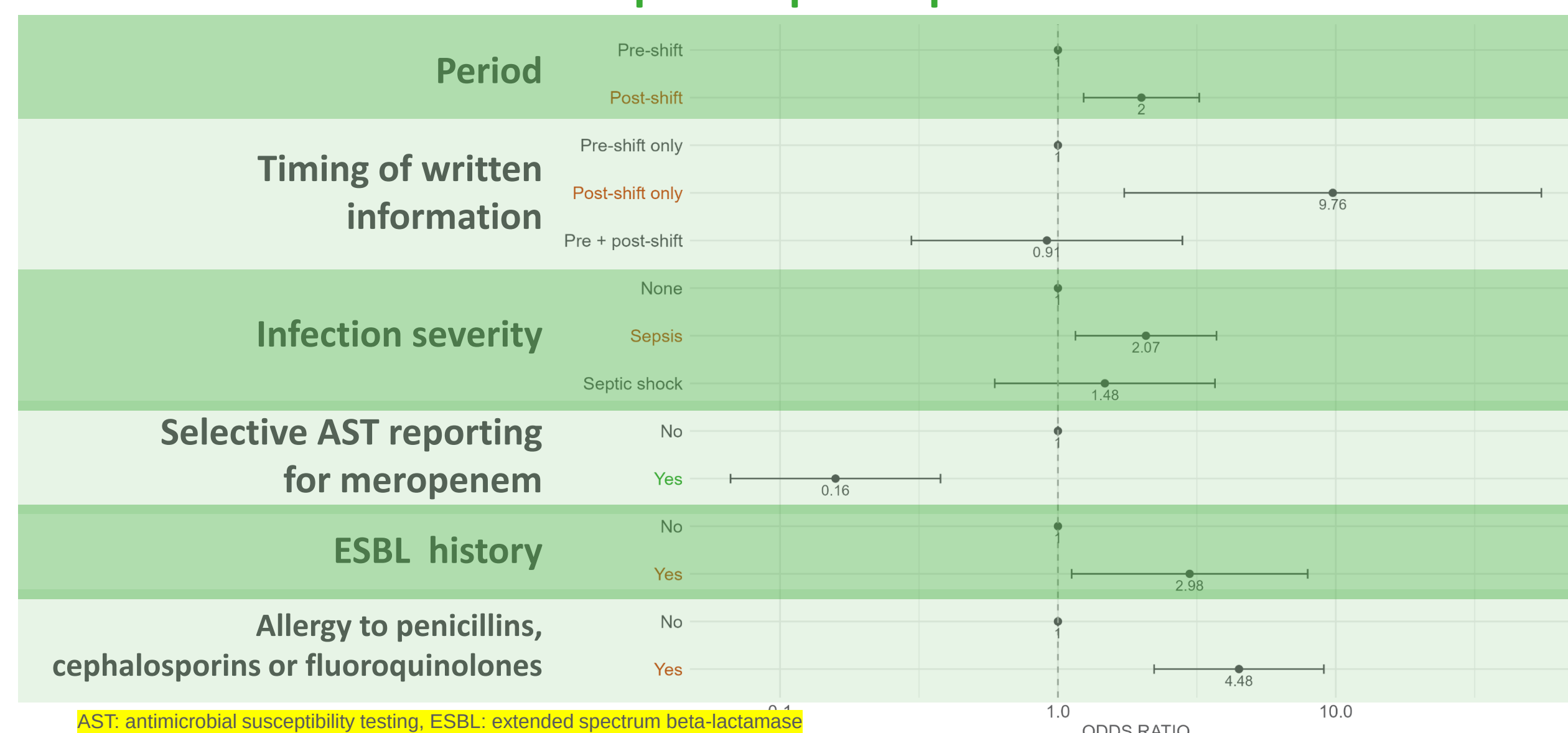
Post-shift

855 patients treated after AST



■ Meropenem ■ Other antibiotics

Factors associated with meropenem prescription



Conclusion

- The 2020 EUCAST criteria led to increased meropenem use for *P. aeruginosa* monobacterial infections in Belgium.
- Key factors influencing meropenem prescriptions:
 - EUCAST shift, delayed information campaigns, sepsis, ESBL history, and allergies to first-line antipseudomonal drugs are associated with increased meropenem use
 - Selective AST reporting may help reduce meropenem use
- Targeted antimicrobial stewardship programs can support appropriate antibiotic use.

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